

IC 161: Applied Electronics

Course Code: IC 161

Course Name: Applied Electronics

L-T-P-C: 3-0-0-3

Prerequisites: None

Intended for: UG (1st Year)

Distribution: Core

Semester: Even/Odd

Approval: 2nd Adhoc Senate; revised in 64th BoA

Course Objective

To understand the principle behind common digital and analog electronics devices. Understanding the fundamentals of electronic components and electrical quantities used in electronic circuits. To analyse basic analog and digital electronic circuits used in engineering applications. To introduce students with the concepts of digital electronics including logic gates, combinational, and sequential circuits.

Course Content

- **Devices and Basic Circuits:** Ideal Diode, PN-Junction Diodes, Clipping and Clamping, Rectification, Zener diode regulator, MOSFET structure and operation, MOSFET switches, biasing, amplifiers (Common Source, Common Gate, Common Drain). Basic logic design with transistors and diodes (CMOS). (10 hours)
- **Introduction to RLC Circuits and Network Theorems:** Fundamentals (Electrical Quantities, Passive Elements, Ohm's Law, Kirchhoff's Laws, Series and Parallel Circuits), Series and Parallel RLC Circuits, Resonance in RLC Circuits, Resonant Frequency, Q-Factor, Bandwidth, Important Network Theorems (Superposition Theorem, Thevenin's Theorem, Norton's Theorem, Maximum Power Transfer Theorem, Reciprocity Theorem, Millman's Theorem). (6 hours)
- **Feedback and Operational Amplifiers:** Introduction to feedback, Operational amplifiers (as a black box), the golden rules, Basic op-amp circuits: Inverting and Noninverting amplifier, Follower, Integrators, Differentiators, Precision rectifiers, Comparators, Schmitt trigger. (7 hours)
- **Digital Electronics:** Number systems (Binary, Decimal, Hexadecimal, Octal), Binary algebra, De-Morgans laws, Combinational Circuits: Adder, Subtractor, Decoder, Encoder, Multiplexers, Demultiplexers Sequential Circuits: Latch, Flipflops, Counters, Shift registers, and Memory. (10 hours)
- **Measurement Transducers:** Temperature, light, acceleration, pressure, force, velocity, magnetic field, particle detectors. (4 hours)
- **Microcontrollers:** Application of Microcontrollers, General Architecture, Interfacing, Bus Signals, Interrupts, Registers, Support Chips. Case study: Compare the architectures of two popularly used microcontrollers, programming of a microcontroller with examples. (5 hours)

Text Books

1. P. Horowitz and Winfield Hill, *The art of electronics*, Cambridge University.



2. M. Mano, *Digital logic design*, Prentice Hall.

Reference Books

1. Adel S. Sedra, Kenneth C. Smith, *Microelectronics Circuits*, Oxford University Press.
2. William Hyat, Jack Kemmerly, Jamie Phillips, Steven Durbin, *Engineering Circuit Analysis*, Mc Graw Hill.